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Code-D Phase-4

Aakash

Medical|IIT-JEE|Foundations

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MM : 720 | **Fortnightly Test for NEET-2027 (XII-Passed_P4)_FT-02D** | **Time : 180 Min**

Topics Covered:
Physics: Motion in a Plane, Laws of Motion, Work, Energy & Power
Chemistry: Structure of Atom, Classification of Elements and Periodicity in Properties
Botany: Cell Cycle & Cell Division, The Living World, Biological Classification-I: (Upto thermoacidophiles)
Zoology: Breathing & Exchange of Gases, Body Fluids & Circulation

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. A vector is not changed if
(1) It is rotated through an arbitrary angle
(2) It is multiplied by an arbitrary scalar
(3) A unit vector is added to the vector itself
(4) It slides parallel to itself

2. Given below are two statements:
Statement I: A vector quantity has both, magnitude and direction and it must follow triangle law of addition.
Statement II: If we multiply a vector $\vec{A}$ with a number λ, then $\left|\lambda\vec{A}\right| = \lambda\left|\vec{A}\right|$.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Statement I is correct but statement II is incorrect
(2) Both statement I and statement II are correct
(3) Both statement I and statement II are incorrect
(4) Statement I is incorrect but statement II is correct

3. If $\vec{A} = 4\hat{i} + 2\hat{j} + 5\hat{k}$ and $\vec{B} = -2\hat{j} - 5\hat{k}$, the angle made by $\vec{A} + \vec{B}$ with positive *y*-axis is
(1) 45° (2) 60°
(3) Zero (4) 90°

4. A particle starts with initial velocity of $\left(\hat{i} - 2\hat{j}\right)\text{m s}^{-1}$ and uniform acceleration $\left(-\hat{i} + 2\hat{j}\right)\text{m s}^{-2}$. The velocity (in m s^{-1}) of the particle at $t = 2$ s, will be
(1) $\hat{i} - 2\hat{j}$ (2) $-\hat{i} + 2\hat{j}$
(3) $-2\hat{i} + \hat{j}$ (4) $-2\hat{i} - \hat{j}$

5. A swimmer can swim with speed of 8 m/s in still water. 800 m wide river is flowing with speed of 4 m/s. Swimmer wants to cross the river in minimum time. Velocity of swimmer with respect to ground is (approximately)
(1) 9 m/s (2) 10 m/s
(3) 12 m/s (4) 5 m/s

6. A particle is projected at angle 60° with horizontal with an initial speed $20\sqrt{2}\text{ m s}^{-1}$. When it makes an angle 45° with horizontal, its speed *v* is
(1) 10 m s^{-1} (2) 20 m s^{-1}
(3) $10\sqrt{2}$ m s^{-1} (4) $10\sqrt{3}$ m s^{-1}

7. Two cars *A* and *B* are moving with equal speed *v*. *A* is moving along north while *B* is moving along east. The magnitude of velocity of *A* with respect to *B*, will be
(1) *v* (2) $v\sqrt{2}$
(3) 2*v* (4) Zero

8. The equation of a projectile is $y = ax - \frac{bx^2}{2}$, its horizontal range is (where *x* is horizontal and *y* is vertical displacement)
(1) $\frac{2a}{b}$ (2) $\frac{a}{b}$
(3) $\frac{a}{2b}$ (4) $\frac{b}{a}$

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89. An isobar of $_{20}Ca^{40}$ is
(1) $_{18}Ar^{40}$
(2) $_{20}Ca^{38}$
(3) $_{20}Ca^{42}$
(4) $_{18}Ar^{38}$

90. In H-atom spectrum electron jumps from 5th excited state to 1st excited state then total number of spectral lines, number of lines in Lyman series and Paschen series respectively are
(1) 10, 4, 3
(2) 15, 0, 4
(3) 15, 4, 5
(4) 10, 0, 3

BOTANY

91. If at the end of meiosis, each daughter cell has four chromosomes, how many chromosomes would a diploid cell have in G_1 phase of cell cycle?
(1) 2
(2) 4
(3) 8
(4) 16

92. Telophase is characterised by
(1) Splitting of centromere
(2) Reappearance of nuclear membrane
(3) Shortening of the spindle fibres
(4) Condensation of chromosomes into chromatin fibres

93. Dissolution of the synaptonemal complex occurs during
(1) Zygotene
(2) Diplotene
(3) Leptotene
(4) Pachytene

94. The microtubules of the spindle fibres attach to the kinetochore during
(1) Anaphase
(2) Prophase
(3) Metaphase
(4) Telophase

95. A bivalent of prophase I is referred as
(1) One centromere with one chromatid
(2) Two centromeres with two chromatids
(3) Two centromeres with four chromatids
(4) Four centromeres with four chromatids

96. In animal cell, centriole duplication occurs during
(1) G_1 phase
(2) S phase
(3) G_2 phase
(4) G_0 phase

97. How many of the following events is/are associated with S phase?
(a) DNA replication
(b) Tubulin protein synthesis
(c) Duplication of mitochondria
(d) Duplication of centrioles
(1) One
(2) Two
(3) Three
(4) Four

98. At the end of meiosis II, from a diploid mother cell
(1) Two haploid cells are formed
(2) Two diploid cells are formed
(3) Four haploid cells are formed
(4) Four diploid cells are formed

99. Which of the following phases is longest in cell cycle?
(1) Prophase
(2) Interphase
(3) Anaphase
(4) Metaphase

100. Select the **incorrect** statement regarding cytokinesis.
(1) In plant cell, it occurs by formation of cell plate
(2) In animal cell, furrow starts at the periphery and then moves inward
(3) Cell plate formation starts from the centre of the cell
(4) In animal cells, cytokinesis is followed by karyokinesis

101. Splitting of centromeres and separation of chromatids occurs at
(1) Telophase
(2) Anaphase
(3) Prophase
(4) Metaphase

102. Some cells in the adult animals that do not appear to exhibit division, e.g., heart cells, exit G_1-phase to enter the stage called
(1) S-phase
(2) M-Phase
(3) Quiescent stage
(4) G_2-phase

103. Read the following **Assertion (A)** and **Reason (R)** and choose the **correct** option.
Assertion (A): To restore the nucleo-cytoplasmic ratio, it is essential for the cell to divide.
Reason (R) : Cell growth disturbs the ratio between the nucleus and cytoplasm.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

104. Interphase is called the resting phase because
(1) It is the most active phase of the cell cycle
(2) There is no apparent activity related to cell division
(3) It does not prepare cell for cell division
(4) It is the phase where cell rests before entering into mitosis

105. Liquid endosperm in coconut exemplifies a structure that is formed when
(1) Karyokinesis is not followed by cytokinesis
(2) Cytokinesis is achieved by cell plate formation
(3) Karyokinesis is delayed
(4) Four haploid uninucleate progenies are formed

106. Read the given statements about the cell cycle and choose the **incorrect** one.
(1) In yeast, the most dramatic period of cell cycle can last less than 4.5 minutes.
(2) Cell cycle is the genetically controlled series of changes that can not be initiated in a newly formed cell.
(3) Completion of DNA synthesis, duplication of cell organelles and synthesis of various proteins in sequential order are the various requirements of a cell which is about to enter M phase.
(4) The duration of cell cycle can vary from organism to organism and also from cell type to cell type.

107. During cell division, the activity of recombinase enzyme is observed in
(1) Leptotene
(2) Zygotene
(3) Pachytene
(4) Diplotene

108. The __A__ together with __B__ forms mitotic apparatus.
Select the option which **correctly** fills the blanks A and B.

	A	B
(1)	Microtubule	One centrosome
(2)	Centriole	Centrosome
(3)	Two asters	Spindle fibres
(4)	One centriole	One aster

(1) (1)
(2) (2)
(3) (3)
(4) (4)

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57. Given below are two statements

Statement I : Ionic radius of F^- is greater than O^{2-}.

Statement II : F^- and O^{2-} are isoelectronic species.

In light of above statements, choose the correct answer in the options given below

(1) Statement I is correct but statement II is incorrect
(2) Both statement I and statement II are correct
(3) Both statement I and statement II are incorrect
(4) Statement I is incorrect but statement II is correct

58. Which of the following is a metalloid?

(1) As (2) Na
(3) S (4) Cl

59. The total number of groups and periods in the modern periodic table respectively are

(1) 7 and 18 (2) 18 and 7
(3) 8 and 12 (4) 12 and 8

60. The correct values of ionisation energies (kJ mol^{-1}) of Si, P, Cl and S are respectively

(1) 786, 1012, 1256, 999 (2) 786, 999, 1012, 1256
(3) 1256, 1012, 999, 786 (4) 786, 999, 1256, 1012

61. The correct increasing order of atomic radii for the following species is

(1) Be < Li < Mg < Na (2) Li < Be < Mg < Na
(3) Li < Be < Na < Mg (4) Be < Li < Na < Mg

62. Which of the following is/are Dobereiner's triad?

(a) Li, Na, K
(b) Cu, Ag, Zn
(c) Fe, Co, Ni
(d) Ca, Sr, Ba

Correct answer is

(1) (a) & (b) only (2) (a) & (d) only
(3) (b) & (c) only (4) (a), (b), (c) & (d)

63. Eka-aluminium is also known as

(1) Gallium (2) Germanium
(3) Calcium (4) Potassium

64. Minimum wavelength of emitted photon of Paschen series for H atom (R_H : Rydberg constant of hydrogen atom) is

(1) $\frac{3}{R_H}$ (2) $\frac{9}{2R_H}$
(3) $\frac{9}{R_H}$ (4) $\frac{1}{R_H}$

65. If the kinetic energy of a particle is tripled, its de-Broglie's wavelength becomes

(1) 2 times (2) 4 times
(3) $\sqrt{2}$ times (4) $\frac{1}{\sqrt{3}}$ times

66. Which of the following is not a subatomic particle?

(1) Electron (2) Neutron
(3) Proton (4) Alpha particle

67. The correct energy order of orbitals for a multielectronic system is

(1) $4f > 5p > 4d > 4p$ (2) $4f > 4d > 5p > 4p$
(3) $5p > 4d > 4p > 4f$ (4) $5p > 4p > 4d > 4f$

68. Which among the following is the incorrect set of quantum numbers?

	n	l	m	s
(1)	4	2	-1	$+\frac{1}{2}$
(2)	5	1	1	$+\frac{1}{2}$
(3)	2	0	0	$+\frac{1}{2}$
(4)	3	2	3	$+\frac{1}{2}$

(1) (1) (2) (2)
(3) (3) (4) (4)

69. Maximum number of electrons in a subshell with n = 4 and l = 3 is

(1) 2 (2) 6
(3) 10 (4) 14

70. Match List-I with List-II and choose the correct code.

	List-I (Orbitals)		List-II (Number of radial nodes)
(a)	2p	(i)	1
(b)	3p	(ii)	3
(c)	3s	(iii)	0
(d)	4s	(iv)	2

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
(2) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

71. Match the following columns I and II.

	Column I (Element)		Column II (Number of unpaired electron(s))
a.	Copper	(i)	5
b.	Manganese	(ii)	4
c.	Chromium	(iii)	1
d.	Iron	(iv)	6

Choose the correct option.

(1) a(ii), b(iv), c(i), d(iii) (2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(i), c(ii), d(iv) (4) a(iii), b(i), c(iv), d(ii)

72. Match List I with List II.

List I (Orbital)	List II (Orbital angular momentum)
a. 4s	(i) $2\sqrt{3}\,\hbar$
b. 3d	(ii) $\sqrt{2}\,\hbar$
c. 2p	(iii) $\sqrt{6}\,\hbar$
d. 4f	(iv) Zero

Choose the correct answer from the options given below.

(1) a(iv), b(ii), c(i), d(i) (2) a(ii), b(iv), c(i), d(iii)
(3) a(iii), b(iv), c(iii), d(i) (4) a(iv), b(ii), c(iii), d(i)

73. Based on the Bohr's model match List I with the list II

	List I		List II
a.	Kinetic energy of the electron in the nth orbit	(i)	$\propto n$
b.	Angular momentum of the electron in the nth orbit	(ii)	$\propto n^2$
c.	Radius of nth orbit	(iii)	$\propto n^{-1}$
d.	Velocity of electron in the nth orbit	(iv)	$\propto n^{-2}$

Choose the correct option

(1) a(iv), b(i), c(ii), d(iii) (2) a(ii), b(i), c(iv), d(iii)
(3) a(i), b(ii), c(iii), d(iv) (4) a(iv), b(iii), c(ii), d(i)

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109. All of the following are significance of mitosis, **except**
(1) It produces genetic variations during sexual reproduction.
(2) It is essential for growth and development of an organism
(3) It helps in healing of wounds
(4) It helps to restore the nucleo-cytoplasmic ratio

110. During prophase
(1) Chromosomes lie at equator of cell
(2) Initiation of condensation of chromosomal material occurs
(3) Chromosomes migrate towards opposite poles
(4) Centromere splits

111. The phase between two successive M phases is called
(1) Prophase (2) Metaphase
(3) Anaphase (4) Interphase

112. Mitosis occurs in all, **except**
(1) Somatic cells (2) Cell lining of gut
(3) Meristematic tissue (4) Pollen mother cell

113. The order of cats and dogs is
(1) Carnivora (2) Mammalia
(3) Primata (4) Diptera

114. According to the rules of binomial nomenclature of organisms
(1) Both generic name and specific epithet are printed in italics
(2) Genus name starts with small letter
(3) Only specific epithet is printed in italics
(4) Specific epithet is written before genus

115. Mark the **incorrect** match w.r.t taxonomic categories of mango.
(1) Kingdom – Angiospermae
(2) Order – Sapindales
(3) Family – Anacardiaceae
(4) Genus – *Mangifera*

116. Find out the **correct descending order** of categories of taxonomical hierarchy.
(1) Species → Genus → Family → Division
(2) Kingdom → Order → Class → Species
(3) Order → Class → Division → Kingdom
(4) Division → Class → Order → Family

117. *Datura* is placed in the family
(1) Poaceae (2) Anacardiaceae
(3) Solanaceae (4) Muscidae

118. The lowest and basic category of classification is
(1) Species (2) Genus
(3) Class (4) Family

119. The processes basic to taxonomy do not include
(1) Characterisation (2) Identification
(3) Nomenclature (4) Phylogeny

120. The word systematics is derived from a _____ word 'systema'.
(1) Greek (2) English
(3) Roman (4) Latin

121. Read the following events
A. Disappearance of nuclear membrane.
B. Two groups of chromosomes get enclosed by separate nuclear membrane.
C. Shortening of microtubules attached to kinetochore aids the movement of sister chromatids towards opposite poles.
D. Microtubules from opposite poles of the spindle gets attached to the kinetochores of the sister chromatids.
Which of the following options represents the correct sequential order of the meiosis II events?
(1) A → D → C → B (2) A → D → B → C
(3) B → A → D → C (4) C → B → A → D

122. The sole members of kingdom Monera
(1) Lack cell walls
(2) Are structurally simple but very complex in behaviour
(3) Have true nucleus
(4) Are photosynthetic due to presence of chloroplast

123. Archaebacteria found in extreme saline conditions are
(1) Thermoacidophiles (2) Methanogens
(3) Mycobacteria (4) Halophiles

124. Match the category of bacteria with the shapes shown by the figure and select the **correct** option.

Column I		Column II
a. Coccus	(i)	
b. Bacillus	(ii)	
c. Vibrium	(iii)	
d. Spirillum	(iv)	

(1) a(iv), b(iii), c(i), d(ii) (2) a(iv), b(i), c(iii), d(ii)
(3) a(ii), b(i), c(iii), d(iv) (4) a(ii), b(iii), c(iv), d(i)

125. __(i)__ are present in the gut of several ruminant animals such as cows and buffaloes and they are responsible for __(ii)__production from the dung of these animals.
Select the **correct** option to fill in the above blanks (i) and (ii) respectively.
(1) Halophiles and Oxygen
(2) Thermophiles and Hydrogen sulphide
(3) Methanogens and Methane
(4) Thermoacidophiles and Sulphuric acid

126. Which of the following characteristics is true for some but not for all the organisms belonging to the Kingdom Monera?
(1) Mode of nutrition – Heterotrophic
(2) Cell type – Prokaryotic
(3) Body organisation – Cellular
(4) Nuclear membrane – Absent

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38. Potential energy associated with a force is given as $U = (x^2 - 5x)$ J (where x is in meter). The magnitude of force at $x = 1$ m will be
(1) 1 N (2) 7 N
(3) 4 N (4) 3 N

39. When a conservative force does positive work on a body
(1) Its potential energy increases
(2) Its potential energy decreases
(3) Its total energy increases
(4) Its total energy decreases

40. A truck moving with velocity 36 km/h is stopped by applying brakes in 2 s. If same truck moves with speed 144 km/h and brakes are applied then the stopping time will be (Assume the same retardation in both cases)
(1) 2 s (2) 5 s
(3) 9 s (4) 8 s

41. Two masses each of 2 kg move towards each other with speed 3 m s^{-1} and 2 m s^{-1}, collide and stick together. The final velocity will be
(1) 1 m s^{-1} (2) 2 m s^{-1}
(3) 2.5 m s^{-1} (4) 0.5 m s^{-1}

42. Coefficient of restitution (e) for perfectly inelastic collision is
(1) 1 (2) Zero
(3) 0.1 (4) Infinite

43. The work done by all the forces (external and internal) on the system equals the change in
(1) Total energy (2) Kinetic energy
(3) Potential energy (4) None of these

44. A vector $\vec{A} = \hat{i} + \sqrt{3}\hat{j}$ is rotated clockwise through an angle 15° about origin in x-y plane. The unit vector along the new vector is
(1) $\sqrt{2}\hat{i} + \sqrt{2}\hat{j}$ (2) $\frac{\hat{i}+\hat{j}}{2}$
(3) $\frac{\hat{i}}{\sqrt{2}} + \frac{\hat{j}}{\sqrt{2}}$ (4) $\frac{1}{2}\hat{i} - \frac{1}{2}\hat{j}$

45. The momentum of a particle varies with time as $P = 2t + 3t^2$ (where P is in kg m/s and t is in seconds). The force acting on particle at $t = 2$ s is
(1) 14 N (2) 12 N
(3) 16 N (4) 20 N

CHEMISTRY

46. The oxidation state and covalency of Al in $[AlCl(H_2O)_5]^{2+}$ respectively are
(1) +2 and 5 (2) +3 and 3
(3) +3 and 5 (4) +3 and 6

47. All the given elements are representative elements except
(1) K (2) Ne
(3) Fe (4) Al

48. The electronic configuration of an element is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^1$. What is the atomic number of the element, which is just above the given element in the periodic table?
(1) 23 (2) 12
(3) 13 (4) 22

49. The element which is in diagonal relationship with Be is
(1) Mg (2) Na
(3) Al (4) B

50. The symbol of element with atomic number 105 is
(1) Uup (2) Unp
(3) Uuu (4) Unn

51. In the modern periodic table, the element with atomic number 33 belongs to
(1) 5th period and 15th group
(2) 3rd period and 13th group
(3) 4th period and 13th group
(4) 4th period and 15th group

52. The element with highest negative electron gain enthalpy will belong to
(1) Group 17, period 3 (2) Group 17, period 2
(3) Group 18, period 2 (4) Group 16, period 2

53. Formation of $O^{2-}(g)$ occurs as follows, comment on the nature of two steps
(i) $O(g) + e^- \rightarrow O^-(g)$
(ii) $O^-(g) + e^- \rightarrow O^{2-}(g)$
(1) Step (i) is exothermic and step (ii) is endothermic
(2) Both the steps are exothermic
(3) Both the steps are endothermic
(4) Step (i) is endothermic and step (ii) is exothermic

54. Match the oxides given in column I with their chemical nature given in column II

Column I	Column II
a. As_2O_3	(i) Basic
b. Cl_2O_7	(ii) Neutral
c. N_2O	(iii) Amphoteric
d. Na_2O	(iv) Acidic

The correct option is
(1) a(i), b(iii), c(iv), d(ii) (2) a(iii), b(iv), c(ii), d(i)
(3) a(iv), b(ii), c(i), d(iii) (4) a(iii), b(i), c(iv), d(ii)

55. Match List-I with List-II.

List-I IUPAC Name	List-II IUPAC Official Name
(a) Ununoctium	(i) Meitnerium
(b) Ununhexium	(ii) Oganesson
(c) Unnilennium	(iii) Lawrencium
(d) Unnilrium	(iv) Livermorium

Identify the correct match.
(1) (a) → (ii), (b) → (iv), (c) → (i), (d) → (iii)
(2) (a) → (ii), (b) → (i), (c) → (iii), (d) → (iv)
(3) (a) → (ii), (b) → (iii), (c) → (iv), (d) → (i)
(4) (a) → (ii), (b) → (i), (c) → (iv), (d) → (iii)

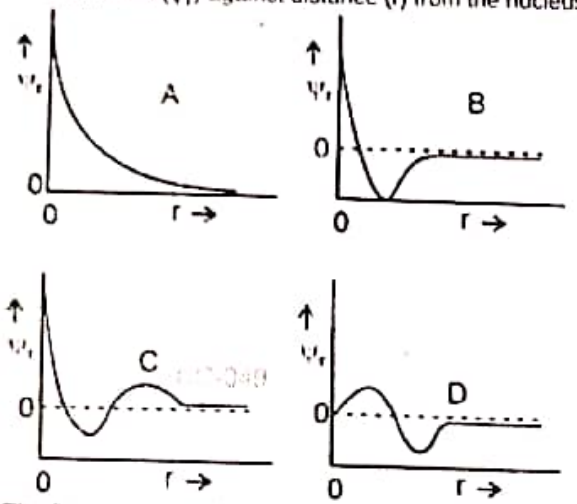
56. The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be
(1) −5.1 eV (2) −10.2 eV
(3) +2.55 eV (4) −2.55 eV

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74. Consider the elements, B, Al, Mg and K, the correct order of their metallic character is
(1) Al > Mg > B > K
(2) B > Al > Mg > K
(3) Mg > Al > K > B
(4) K > Mg > Al > B

75. If a 30 watt bulb emits monochromatic light of wavelength 1320 nm, then the number of photons emitted per second by the bulb will be (Given : $h = 6.6 \times 10^{-34}$ Js)
(1) 4×10^{21}
(2) 2×10^{20}
(3) 2×10^{22}
(4) 4×10^{19}

76. Consider the following schematic plots of orbital wavefunction (ψ_r) against distance (r) from the nucleus.



The figure representing two radial nodes in the orbital is
(1) C
(2) D
(3) A
(4) B

77. The lightest particle among the following is
(1) Electron
(2) Proton
(3) Neutron
(4) α-particle

78. Given below are two statements:
Statement-I: For hydrogen atom n = 1, the electron has a more negative energy than n = 5, which means that the electron is more loosely bound in n = 1 orbit
Statement-II: Maximum number of electrons that a subshell can accommodate is $4l + 2$.
In the light of above statements, choose the correct answer from the options given below.
(1) Both statements I and II are correct
(2) Statement I is incorrect but statement II is correct
(3) Statement I is correct but statement II is incorrect
(4) Both statements I and II are incorrect

79. If the uncertainty in momentum of an electron is 5×10^{-6} kg ms^{-1} then the uncertainty in its position will be ($h = 6.626 \times 10^{-34}$ kg $m^2 s^{-1}$)
(1) 1.05×10^{-29} m
(2) 2.5×10^{-30} m
(3) 3.14×10^{-29} m
(4) 4.5×10^{-30} m

80. Ratio of potential energy associated with the fourth orbit of He^+ ion to that of second orbit of Be^{3+} ion is
(1) 2 : 3
(2) 16 : 1
(3) 3 : 5
(4) 1 : 16

81. Consider the following statements:
(i) Paschen, Brackett and Pfund series of spectral lines for hydrogen atom appear in the infrared region of the electromagnetic spectrum.
(ii) The value of wave function depends upon the coordinates of the electron in the atom.
(iii) The probability of finding an electron at a point within an atom is proportional to the square of the orbital wave functional, $|\psi|^2$ at that point.
(iv) Two electrons in an orbital cannot have same set of first three quantum numbers (*i.e.*, n, l and m_l).
Choose the set containing all correct statements:
(1) (i), (iii) and (iv) only
(2) (ii) and (iv) only
(3) (i), (ii) and (iii) only
(4) (i) and (iv) only

82. Wavelength observed in visible region of hydrogen spectrum is expressed as $\lambda = \frac{Kn^2}{n^2-4}$. The value of K in terms of Rydberg constant(R) is
(1) $\frac{4}{R}$
(2) $\frac{R}{4}$
(3) $\frac{5}{2}R$
(4) R

83. The maximum kinetic energy of photoelectrons emitted when a light of frequency 1×10^{16} Hz is irradiated on a metal surface with threshold frequency (ν_0) = 2×10^{15} Hz is
(1) 3.6×10^{-18} J
(2) 1.7×10^{-20} J
(3) 7.5×10^{-19} J
(4) 5.3×10^{-18} J

84. The correct order of wavelength of given electromagnetic waves is
(1) Microwave > UV > IR > Radio waves
(2) Radiowaves > Microwave > UV > IR
(3) Microwave > Radiowave > IR > UV
(4) Radiowaves > Microwave > IR > UV

85. Calculate the wave number (m^{-1}) if, the frequency of violet radiation is 7.5×10^{14} Hz
(1) 2.5×10^6
(2) 4×10^{-7}
(3) 1.33×10^{-15}
(4) 2.5×10^7

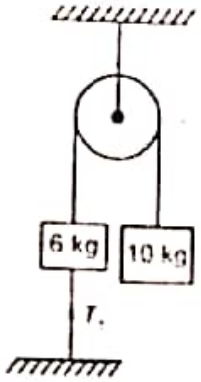
86. Which experimental phenomenon is explained by wave nature of electromagnetic radiation?
(1) Diffraction
(2) Black-body radiation
(3) Line spectra of hydrogen atom
(4) Photoelectric effect

87. Rutherford's experiment, which established the nuclear model of the atom, used a beam of
(1) Helium atoms, which impinged on a metal foil and got scattered
(2) β-particles, which impinged on a metal foil and got absorbed
(3) γ-rays which impinged on a metal foil and ejected electrons
(4) Helium nuclei, which impinged on a metal foil and got scattered

88. Which of the following quantum number specifies the size of the shell?
(1) Principal quantum number
(2) Azimuthal quantum number
(3) Magnetic orbital quantum number
(4) Spin quantum number

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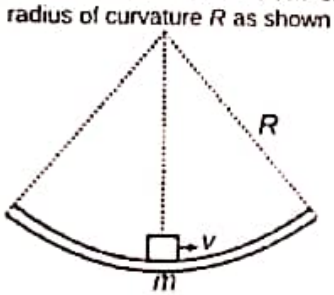
23. Two bodies of masses 10 kg and 6 kg are attached to the ends of a string passing over a pulley. The 6 kg mass is attached to the table top by another string. The tension T_1 in the string is equal to



(1) 100 N (2) 60 N
(3) 40 N (4) 20 N

24. A bullet is fired with a speed 200 m/s. If the mass of bullet is 100 gram and of gun is 2 kg, then the recoil speed of gun is
(1) 20 m/s (2) 2 m/s
(3) 10 m/s (4) 5 m/s

25. Normal reaction of a block on a vertical curved track of radius of curvature R as shown in figure is



(1) $N = mg - \frac{mv^2}{R}$ (2) $N = mg + \frac{mv^2}{R}$
(3) $N = \frac{mv^2}{R} - mg$ (4) Zero

26. Sand is poured vertically on a conveyor belt at the rate of 2 kg/s. If belt is moving horizontally with velocity 4 m/s, then additional force required by engine to keep the belt moving with same constant velocity is
(1) 4 N (2) 8 N
(3) 16 N (4) 32 N

27. If no external force acts on a system, then consider the following statements
(a) acceleration of centre of mass is zero
(b) centre of mass of system may be at rest or moving with constant velocity
(c) momentum of system is changing with time
(d) momentum of system remains constant.
Choose the correct answer from the options given below.
(1) (a), (b) and (d) are true
(2) (a), (c) and (d) are true
(3) (b), (c) and (d) are true
(4) (a), (b), (c) and (d) all are correct

28. Area under force-time curve is equal to
(1) Change in linear momentum
(2) Impulse
(3) Work done
(4) Both (1) and (2)

29. Given below are two statements labelled as Assertion (A) and Reason (R)
Assertion (A): If a body is in equilibrium while suspended vertically with the help of a string then tension in the string balances the weight of the body.
Reason (R): Static friction force is a self adjusting force.
Select the most appropriate answer from the options given below
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false and (R) is also false.

30. If the numerical value of kinetic energy of a body is 4 times the numerical value of its momentum, then velocity of the body is
(1) 4 unit (2) 8 unit
(3) 2 unit (4) 6 unit

31. If the force on a particle depends on position x as $F = (5x)$ N, then work done on the particle in moving from $x = 0$ to $x = 2$ m is
(1) 5 J (2) 10 J
(3) 15 J (4) 20 J

32. A ball is dropped on the ground from a height of 100 m. If coefficient of restitution is 0.6, then height to which the ball will rebound is
(1) 60 m (2) 40 m
(3) 36 m (4) 64 m

33. How much oil in kg, a 100% efficient pump of 5 kW power can raise to a tower 50 m high every hour? (g = 10 m/s^2)
(1) 7200 kg (2) 18000 kg
(3) 36000 kg (4) 72000 kg

34. A ball of mass 100 g is thrown in vertically upward direction with velocity 20 m/s. The increase in potential energy of the ball when it reaches at the maximum height is (take g = 10 m/s^2)
(1) 1 J (2) 20 J
(3) 10 J (4) 100 J

35. Two springs of spring constants 1000 N m^{-1} and 2000 N m^{-1} are stretched with same force. They have stored potential energy in the ratio of
(1) 2 : 1 (2) 1 : $\sqrt{2}$
(3) 4 : 1 (4) 1 : 4

36. A block of mass m is attached to one end of string and other end is fixed. It is executing vertical circular motion. For just completing the loop, match the column-I with column-II. (Symbol have usual meanings)

Column-I	Column-II
A. Speed at bottom-most point	P. $\sqrt{3gl}$
B. Speed at top-most point	Q. $\sqrt{5gl}$
C. Speed at horizontal position of string	R. $\sqrt{gl}$

(1) A-P, B-Q, C-R (2) A-P, B-R, C-Q
(3) A-R, B-Q, C-P (4) A-Q, B-R, C-P

37. Which of the following forces is non-conservative in nature?
(1) Gravitational force (2) Spring force
(3) Electrostatic force (4) Frictional force

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127. In two kingdom system of classification, algae and fungi are placed in the same kingdom. This inclusion indicates that the two kingdom system does not distinguish between
(1) Unicellular and multicellular organisms
(2) Eukaryotic and prokaryotic organisms
(3) Photosynthetic and non photosynthetic organisms
(4) Cellular and acellular organism

128. Kingdom Protista brought together *Chlamydomonas* and *Chlorella* with *Paramoecium* and *Amoeba* which were earlier placed in the kingdoms
(1) Monera and Animalia
(2) Plantae and Monera
(3) Plantae and Animalia
(4) Monera and Fungi

129. The most abundant micro-organisms are
(1) Bacteria
(2) Mycoplasma
(3) Viroids
(4) Prions

130. Earliest scientific attempt for classification of organisms was performed by
(1) R. H. Whittaker
(2) Carl Woese
(3) Linnaeus
(4) Aristotle

131. In five kingdom classification, how many kingdoms may have photosynthetic members but without membrane bound cell organelles?
(1) Three
(2) Four
(3) One
(4) Two

132. In five kingdom classification, only single celled eukaryotes are included in
(1) Monera
(2) Protista
(3) Fungi
(4) Plantae

133. In the three-domain system, Monera can be divided into
(1) One domain
(2) Three domains
(3) Two domains
(4) Five kingdoms

134. The vast majority of bacteria
(1) Can synthesize their own food from inorganic substances
(2) Use sunlight to synthesize their food
(3) Are heterotrophic in nature
(4) Are complex in structure but simple in behaviour

135. In the kingdom system of classification proposed by Linnaeus, fungi were placed in the group
(1) Plantae
(2) Animalia
(3) Mycota
(4) Monera

ZOOLOGY

136. Among the following, which blood vessel carries deoxygenated blood?
(1) Pulmonary artery
(2) Pulmonary vein
(3) Aorta
(4) Hepatic artery

137. The opening between right atrium and right ventricle is guarded by
(1) Tricuspid valve
(2) Mitral valve
(3) Bicuspid valve
(4) Semilunar valve

138. Pacemaker of the human heart is located in the
(1) Cardiac centre of medulla oblongata
(2) Upper right corner of the right atrium of heart
(3) Lower left corner of the right atrium of heart
(4) Inter-ventricular septum of heart

139. Oxygen is transported to tissues of the body through
(1) RBCs
(2) WBCs
(3) RBCs and WBCs
(4) RBCs and hormones

140. Select the correct option to complete the analogy w.r.t. respiratory structures.
Terrestrial arthropods : Tracheal tubes :: Aquatic arthropods : _____
(1) Moist cuticle
(2) Gills
(3) Lungs
(4) Green glands

141. In earthworms, respiration takes place through
(1) Moist skin
(2) Gills
(3) Lungs
(4) Tracheal system

142. The rate of breathing in a healthy adult human under normal physiological condition is
(1) 20-25 times/minute
(2) 12-16 times/minute
(3) 12-16 times/second
(4) 40-45 times/hour

143. Prothrombin is converted to thrombin to attain hemostasis in the presence of
(1) Fibrin
(2) Fibrinogen
(3) Thrombokinase
(4) Sodium ions

144. Which of the following prevents collapsing of trachea?
(1) Muscles
(2) Diaphragm
(3) Ribs
(4) Cartilaginous rings

145. Maximum amount of O_2 that can be delivered to the tissues by 500 mL of blood during strenuous exercise is about
(1) 75 mL
(2) 100 mL
(3) 25 mL
(4) 1.34 mL

146. During a cardiac cycle, each ventricle pumps out about 70 mL of blood, which is called
(1) End systolic volume
(2) Stroke volume
(3) End diastolic volume
(4) Cardiac output

147. The second heart sound can be heard when
(1) Atrio-ventricular valves open
(2) Semilunar valves open
(3) Atrio-ventricular valves close
(4) Semi-lunar valves close

148. Partial pressure of oxygen in systemic arteries is
(1) 40 mm Hg
(2) 45 mm Hg
(3) 104 mm Hg
(4) 95 mm Hg

149. Closed circulatory system is present in all
(1) Chordates
(2) Non-chordates
(3) Vertebrates
(4) Invertebrates

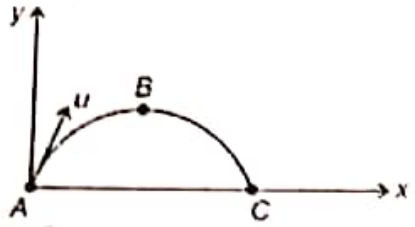
150. The blood clot is formed mainly by
(1) Thrombocytes only
(2) Thrombokinase only
(3) Thrombins and Fibrinogen
(4) Fibrins and formed elements

151. **Assertion (A)** : Coronary artery disease (CAD) is often referred to as atherosclerosis.
Reason (R) : It is caused due to deposition of cholesterol, fibrous tissue, *etc*, into pulmonary artery
In the light of above statements, choose the correct option.
(1) Both (A) and (R) are false
(2) (A) is true but (R) is false
(3) Both (A) and (R) are true and (R) explains (A) correctly
(4) Both (A) and (R) are true but (R) does not explain (A) correctly

152. A special fibrous cord that prevents the prolapse of AV valves into atria is called
(1) [illegible] endinae
(2) Papillary muscles
(3) [illegible]
(4) [illegible]

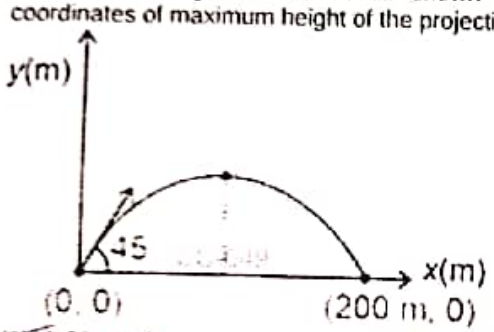
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9. The velocity of a projectile at the initial point A is $(\hat{i}+\hat{j})$ m/s. Ratio of speed at point B and point C is



(1) 1
(2) $\sqrt{2}$
(3) $\frac{1}{\sqrt{2}}$
(4) $\frac{1}{2}$

10. A particle is projected at an angle of 45° such that it achieves a range of 200 m as shown in figure. The coordinates of maximum height of the projectile in meter is?



(1) (100, 100)
(2) (100, 50)
(3) (50, 100)
(4) (50, 50)

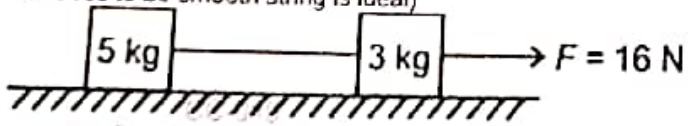
11. Let a_r and a_t represent radial and tangential acceleration. The motion of a particle may be circular if
(1) $a_r = 0, a_t \neq 0$
(2) $a_r \neq 0, a_t = 0$
(3) $a_r \neq 0, a_t \neq 0$
(4) Both (2) and (3)

12. A body performing uniform circular motion completed 140 revolutions in a second. Its angular speed is
(1) 880 rad/s
(2) 440 rad/s
(3) 220 rad/s
(4) 240 rad/s

13. Two particles are moving in circular paths of same radii with angular speeds 2 rad/s and 4 rad/s. The ratio of their centripetal accelerations is
(1) 1 : 4
(2) 1 : 2
(3) 1 : 1
(4) 1 : 3

14. Speed at maximum height of a projectile is $\frac{1}{2}$ of its initial speed. Its angle of projection from the horizontal is
(1) 60°
(2) 30°
(3) 45°
(4) 37°

15. What is the acceleration of 5 kg mass? (Assume all the surfaces to be smooth string is ideal)



(1) 2 m/s^2
(2) 4 m/s^2
(3) 1 m/s^2
(4) 3 m/s^2

16. If a body is acted upon by external unbalanced forces, then
(1) It will always be at rest
(2) It will always move with a constant velocity
(3) It will move with some acceleration
(4) Both (1) and (2)

17. If a body under the action of force $\vec{F} = (3\hat{i}+4\hat{j}-5\hat{k})$ N, acquires an acceleration of magnitude $\sqrt{2}$ m/s^2, then mass of body is
(1) 5 kg
(2) 2 kg
(3) 10 kg
(4) $5\sqrt{2}$ kg

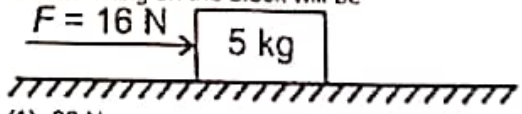
18. A body of mass M hits normally a rigid wall with speed V and bounces back with same speed. The impulse imparted to wall is
(1) MV
(2) $\frac{MV}{2}$
(3) $2MV$
(4) Zero

19. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**
Assertion (A): Even if the net external force acting a body is zero, its linear momentum may not necessarily be conserved.
Reason (R): The internal forces between the particles of a body can change the linear momentum of the body.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is false but (R) is true
(4) Both (A) and (R) are false

20. If the radius of bend is 13 m and coefficient of friction between the road and the tyres is 0.5 Maximum speed with which a car can turn on a bend without skidding is about to ($g = 10$ m/s^2)
(1) 8 m/s
(2) 7 m/s
(3) 10 m/s
(4) 5 m/s

21. A boy of mass 40 kg climbs up a rope with an acceleration of 2 m/s^2, the tension in the rope is ($g = 9.8$ m/s^2)
(1) 472 N
(2) 435 N
(3) 232 N
(4) 568 N

22. A block of mass $m = 5$ kg is placed on a rough horizontal surface having friction coefficient $\mu_s = 0.4$. If a force $F = 16$ N is applied on the block as shown in the figure, then value of friction acting on the block will be



(1) 20 N
(2) 16 N
(3) 8 N
(4) Zero

153. In humans, inspiration is possible if the pressure within the lungs is
(1) Less than the atmospheric pressure
(2) More than the atmospheric pressure
(3) Equal to the atmospheric pressure
(4) Double than the atmospheric pressure

154. Blood is a specialized connective tissue devoid of
(1) Matrix
(2) Fibres
(3) Proteins
(4) Formed elements

155. Select the incorrect option w.r.t. normal expiration.
(1) Volume of thoracic cavity decreases
(2) Decrease in pulmonary volume
(3) Ribs and sternum are raised
(4) Diaphragm relaxes

156. Volume of air inspired or expired during a normal respiration is about
(1) 1100 mL
(2) 500 mL
(3) 2500 mL
(4) 1200 mL

157. Arteries differ from veins as former have
(1) No valves
(2) Slower flow of blood under low pressure
(3) Thin and collapsible walls
(4) Fewer muscle fibres

158. The tunica externa layer of blood vessels
(1) Does not contain collagen fibres
(2) Consists of smooth muscle
(3) Is in direct contact with endothelium lining
(4) Lacks elastic fibers and an endothelium

159. Partial pressure of oxygen in deoxygenated blood is similar to
(1) pCO_2 in tissue
(2) pO_2 in alveoli
(3) pCO_2 in oxygenated blood
(4) pCO_2 in deoxygenated blood

160. There is a right shift of the O_2 dissociation curve under all of the following conditions, **except**
(1) High pCO_2
(2) High H^+
(3) High temperature
(4) High pH

161. Select the human organ often referred to as 'graveyard' of RBCs.
(1) Thymus
(2) Bone marrow
(3) Liver
(4) Spleen

162. Which of the following types of WBCs release heparin and are involved in inflammatory reactions?
(1) Lymphocytes
(2) Eosinophils
(3) Basophils
(4) Neutrophils

163. Complete the analogy by selecting the correct option.
Respiratory rhythm centre : Medulla oblongata : : Pneumotaxic centre : _____.
(1) Medulla oblongata
(2) Cerebrum
(3) Cerebellum
(4) Pons

164. A person encountered an allergen due to which he had difficulty in breathing. He went to a hospital and doctor observed wheezing alongwith redness and swelling in bronchi and bronchioles.
Identify the disorder and choose the **correct** option.
(1) Asthma
(2) Silicosis
(3) Emphysema
(4) Anthracis

165. Exchange of gases takes place in which part of the respiratory system?
(1) Alveoli
(2) Tertiary bronchi
(3) Larynx
(4) Nose

166. Part of respiratory tract **not** supported by incomplete cartilaginous rings is
(1) Trachea
(2) Tertiary bronchi
(3) Initial bronchioles
(4) Alveoli

167. Systolic blood pressure of a healthy adult human male, under normal physiological conditions, is
(1) 120 mm Hg
(2) 80 mm Hg
(3) 200 mm Hg
(4) 50 mm Hg

168. The component of blood completely absent in lymph are
(1) WBCs
(2) Clotting factors
(3) Plasma proteins
(4) RBCs

169. Trachea divides into bronchi at the level of which thoracic vertebra ?
(1) 5th
(2) 7th
(3) 3rd
(4) 1st

170. If the cardiac output of a person is about 6.3 Litres/minute and his end systolic and end diastolic volumes of the left ventricle are 60 mL and 130 mL, respectively, then heart rate of the person will be
(1) 80 beats per minute
(2) 72 beats per minute
(3) 90 beats per minute
(4) 75 beats per minute

171. The average life span of an adult human RBC is _____ and the organ considered as the graveyard of erythrocytes is _____. Choose the option that fills the respective blanks correctly.
(1) 10 days; spleen
(2) 120 days; liver
(3) 120 days; spleen
(4) 180 days; liver

172. In human heart, a thin muscular wall separates
(1) Right and left ventricles
(2) Right auricle from right ventricle
(3) Right and left atria
(4) Left auricle from left ventricle

173. Neutrophils and monocytes differ from each other in all of the following, **except**
(1) The shape of their nuclei in mature form
(2) Their contribution to total WBC content of blood
(3) Performing phagocytosis of foreign organisms
(4) Their cytoplasmic contents

174. Which ions play an important role in blood clotting?
(1) Phosphate
(2) Magnesium
(3) Sodium
(4) Calcium

175. How many of the below mentioned structures are the components of human respiratory system?

Larynx, Green glands, bronchi, bronchioles, tracheoles, nasopharynx

(1) Four
(2) Three
(3) Two
(4) Five

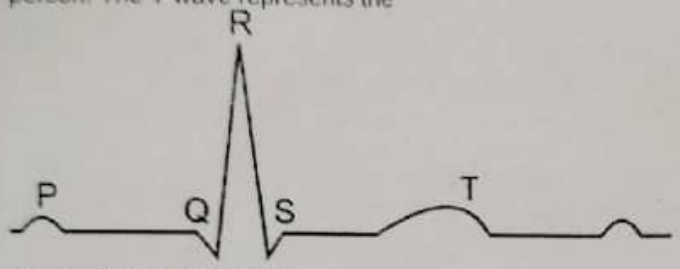
176. Megakaryocytes are present in
(1) Thymus
(2) Heart
(3) Bone marrow
(4) Liver

177. Tracheal tubes are respiratory structures of
(1) Insects
(2) Molluscs
(3) Coelenterates
(4) Annelids

178. The most abundant cells among WBCs are
(1) Basophils
(2) Neutrophils
(3) Monocytes
(4) Eosinophils

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179. The diagram given below is the standard ECG of a normal person. The T-wave represents the



(1) Excitation of the atria
(2) Depolarisation of the ventricles
(3) Repolarisation of the ventricles
(4) End of ventricular diastole

180. Select the **correct** option w.r.t. the features of human heart.
(1) Endodermal in origin
(2) Functions can be moderated by endocrine system
(3) Slightly tilted towards the right side of the body
(4) Protected by a single layer of membranous pericardium

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